

PRODUCT REQUIREMENTS DOCUMENT

MOTS Shipt

Agentic load booking for U.S. freight brokers

Parallel carrier rate negotiation • Automated booking • Compliance verification • Exception orchestration

Product intent Convert a broker-approved load and policy into a compliant carrier commitment with minimal human touch, while preserving control, auditability, and fast exception recovery.

Field	Value
Document status	Draft v1.0 for discovery and design validation
Product owner	Technical Product Management
Target segment	U.S. freight brokers managing thousands of loads per month
Primary users	Supply Chain Strategic Operations Manager (Admin), carrier sales representatives, carrier dispatchers, compliance teams
Planning horizon	MVP in 16 weeks; controlled scale-up over the following 2 quarters
Last updated	September 12, 2026

Decision note

All numerical targets, cost ranges, pricing, and market scenarios in this PRD are planning hypotheses unless a source is cited. Baselines will be established during discovery and the first four weeks of shadow mode.

Executive Summary

MOTS ShipIt is an agentic workflow platform that ingests an approved load, identifies eligible carriers, contacts several carriers in parallel, negotiates inside broker-defined commercial guardrails, verifies carrier identity and compliance, books the selected carrier, and records every decision in the broker's transportation management system (TMS).

The product is designed for the operational reality of freight brokerage: noisy phone calls, fragmented systems, inconsistent carrier data, time-sensitive pricing, possible fraud, and frequent exceptions. A policy engine owns hard constraints; AI agents interpret unstructured interactions and coordinate tools; humans retain authority over defined risk and value thresholds.

MVP outcome

North-star target At least 60% of eligible loads booked without a human touch, with 100% hard-gate compliance and no statistically significant degradation in gross margin or service outcomes versus the broker's control workflow.

Product principles

- Policy before autonomy: deterministic rules control eligibility, rate ceilings, identity, authority, insurance, and commitment.
- Evidence before action: every recommendation and booking must link to source data, tool results, and an immutable event log.
- Escalate uncertainty: low-confidence intent, conflicting terms, suspected fraud, and novel exceptions route to a person.
- Parallelize safely: agents contact multiple carriers concurrently but maintain a single authoritative load state and atomic commitment.
- Learn from outcomes, not raw acceptance: feedback includes tender acceptance, pickup, delivery, claims, margin, and human corrections.

Scope at a glance

Phase	In scope for MVP	Out of scope for MVP
This is MVP for phase 1	Dry van and reefer truckload in the contiguous U.S.; repeat and approved carriers; business-hours and approved after-hours outreach.	Hazmat, oversize, cross-border customs, drayage, household goods, and regulated special commodities.
Deferred to Phase 2	Voice, SMS, and email outreach; TMS load intake and update; rate negotiation; compliance hard gates; rate confirmation.	Autonomous shipper pricing, load creation, collections, claims adjudication, or carrier payment.
Descope	English-first; Spanish handoff or controlled scripts only after evaluation.	Open-domain conversation, general dispatch services, or unrestricted autonomous contracting.

Key decisions required

Decision	Owner	Needed by
Pilot customer, TMS, lanes, equipment types, and eligible carrier pool	Product + Customer	Week 2
Approved contact basis, AI disclosure, recording consent, and retention policy	Legal + Compliance	Before live calls
Rate guardrails and exception authority by customer and load type	Customer Ops + Finance	Before shadow mode
Identity verification and carrier fraud controls	Security + Compliance	Before booking writes

WEEK 1

Problem & Solution Definition

PROBLEM DEFINITION

What problem is this solving?

For each time-sensitive load, a broker must locate suitable capacity, repeatedly contact carriers, interpret responses across noisy calls and messages, negotiate a viable rate, verify that the carrier is legitimate and compliant, resolve exceptions, secure acceptance, and update the TMS. The workflow is repetitive but not simple: context is scattered, terms are conversational, risk is asymmetric, and the first acceptable answer is not always the best operational choice.

Job to be done

When an approved load needs coverage, help the brokerage find, verify, negotiate with, and commit the best eligible carrier quickly, so the load is covered within policy without requiring a representative to coordinate every interaction.

Who are we solving for?

Persona	Need	Current pain	Success signal
Supply Chain Strategic Operations Manager (Admin)	Full authorized control of brokerage operations, policies, users, integrations, automation, and exceptions	Fragmented operational visibility and limited ability to control workflows end to end	Can configure, monitor, intervene in, and audit all operations from one control plane
Carrier sales representative	Cover loads quickly and protect margin	Repetitive outreach, context switching, manual note-taking	Works only high-value or ambiguous exceptions
Operations manager	Predictable coverage and service	Late escalations and inconsistent process execution	Coverage SLA, queue health, and interventions are visible
Compliance specialist	Prevent fraud and unsafe/noncompliant bookings	Multiple portals and stale snapshots	Every booking has a time-stamped evidence bundle
Brokerage customer / shipper	Reliable pickup at an acceptable cost	Uncertain status and variable communication	Fewer missed pickups and faster confirmed capacity
Carrier dispatcher	Clear load details and fast agreement	Repeated calls, missing terms, delayed confirmations	One accurate conversation and prompt confirmation

Admin authority

Supply Chain Strategic Operations Manager The Admin can perform all authorized operational activities: manage users and roles; configure broker, customer, lane, carrier, contact, and integration settings; create and version policies; set rate and autonomy thresholds; start, pause, resume, reassign, approve, override, or cancel workflows; manage exceptions; review conversations and evidence; reconcile bookings; and access operational, compliance, and financial reporting. Mandatory legal, compliance, security, segregation-of-duties, and audit controls remain enforced.

Why is this problem worth solving?

Freight is economically significant and operationally time-sensitive. The Bureau of Transportation Statistics reports that the U.S. transportation system moved an average of about 55.5 million tons of freight worth more than \$51.2 billion per day in 2023 [R1]. The product opportunity is narrower: remove coordination work from high-volume broker workflows while improving control over carrier selection and documentation.

Differentiated product thesis

- Closed-loop execution: Shipt does not stop at a suggestion or transcript; it completes verified workflow steps through TMS and compliance tools.
- Multi-agent concurrency: it negotiates with several eligible carriers while a load-state coordinator prevents double booking.
- Broker-owned policy: commercial and compliance rules are versioned, testable, and enforced outside the language model.
- Operational learning: outcomes and human corrections improve ranking, prompts, and policies without allowing uncontrolled self-modification.
- Exception-first design: ambiguity, risk, and failure are visible work queues rather than hidden model behavior.

Why Agentic AI?

Capability	Best-fit method	Why
Authority, insurance, contact permission, rate ceiling	Deterministic policy and tool checks	Binary or computable constraints must be reproducible and auditable.
Understand speech, interruptions, accents, and free-form messages	Speech models + LLM extraction	Inputs are unstructured and cannot be fully enumerated.
Choose next action and channel	Bounded agent planner	The workflow changes based on carrier response, elapsed time, and load state.
Rank candidate carriers and expected acceptance	Statistical model	Historical lane, carrier, rate, and service signals support prediction.
Commit a booking	Transactional service with policy recheck	Requires atomic state transition; the LLM never writes directly.

A rules-only system cannot reliably interpret open-ended conversations or adapt the sequence of outreach. A standalone LLM is also insufficient because it may be inconsistent, cannot be trusted as the

system of record, and must not own irreversible transactions. Shipt therefore combines probabilistic AI with deterministic services, tool permissions, and human approval.

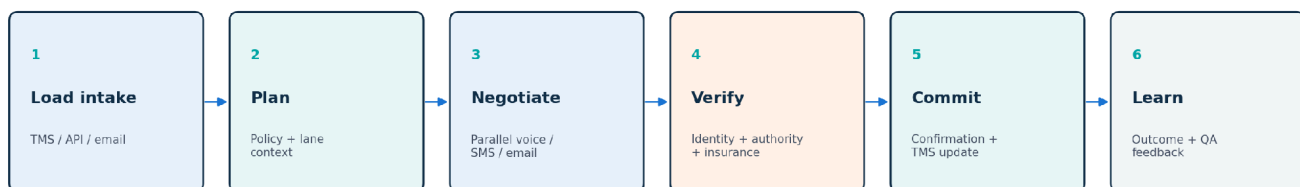
How will we know the problem is solved?

Metric	Definition	MVP target	Guardrail
North star: compliant touchless booking rate	Eligible loads booked with no human action and all hard checks passed	≥60% by week 8 of pilot	No bypass of required checks
Median time to cover	Load ready-to-source → carrier commitment	≥50% faster than baseline	P90 does not worsen
Human minutes per eligible load	Active employee handling time	≥65% reduction	Rework included
Rate performance	Booked linehaul vs approved benchmark / ceiling	Non-inferior to control; target ≥2% improvement	Never exceed ceiling without approval
Booking accuracy	Confirmed terms match TMS and rate confirmation	≥99.5%	100% for critical fields
Compliance gate precision	Blocked bookings that should be blocked	≥99%	Zero known hard-gate bypasses
Carrier experience	Post-interaction CSAT / opt-out rate	≥4.2/5; opt-out <2%	No repeat contact after opt-out
Service outcome	Pickup/delivery failure attributable to carrier selection	No degradation vs control	Auto-pause on threshold breach

Eligibility denominator: loads that are in-scope, have complete data, fall within approved lanes/equipment, have a usable candidate pool, and do not trigger a pre-sourcing exception. Report both overall automation and eligible-load automation to prevent denominator gaming.

SOLUTION DEFINITION

User flow



Human escalation can occur after any step; no booking is committed unless hard policy checks pass.

Happy-path workflow

1. The TMS publishes a sourcing-ready load with lane, equipment, schedule, commodity, weight, customer rules, target rate, and hard ceiling.
2. The orchestrator validates completeness, locks the load version, applies policy, and creates an eligible candidate list.
3. Carrier agents contact a bounded batch in parallel using approved channels, identify the AI system, verify the person's authority, and confirm critical load facts.

4. Each agent extracts an offer with confidence and explicit evidence, then negotiates within the approved range and concession policy.
5. The selection service scores viable offers using rate, service, fit, compliance freshness, and customer preference. It selects one only after rechecking load state and compliance.
6. The booking service atomically commits the carrier, closes other conversations, generates the rate confirmation, writes the TMS, and sends notifications.
7. All artifacts, outcomes, and human corrections enter the audit and learning pipeline.

Exception flows

Trigger	System action	Human action / SLA
Low-confidence speech or conflicting terms	Read back terms; try alternate channel; freeze commitment	Review transcript/audio within 5 min
Rate exceeds ceiling or carrier requests new term	Stop negotiation; show offer and context	Approve, counter, or reject
Carrier identity or payment change risk	Block booking; preserve evidence	Compliance investigation
Load changes during negotiation	Invalidate offers affected by changed fields; notify agents	Review if a carrier already accepted
Preferred carrier fails to confirm	Expire reservation and move to next ranked offer	Intervene only after policy timeout
Tool outage or stale compliance data	Fail closed for booking; continue read-only outreach only if policy allows	Use documented manual process

Functional requirements

ID	Capability	Requirement	Priority
FR-01	Load intake	Ingest sourcing-ready loads and change events through TMS API/webhook; validate required fields and version each change.	Must
FR-02	Policy configuration	Allow authorized admins to define eligible lanes, equipment, carrier tiers, contact windows, rate bounds, concessions, and approval thresholds.	Must
FR-03	Candidate generation	Generate a deduplicated, ranked carrier list from approved sources with reason codes and exclusion explanations.	Must
FR-04	Parallel outreach	Contact a configurable number of carriers concurrently across voice, SMS, and email while respecting frequency caps and opt-outs.	Must
FR-05	Voice interaction	Handle voicemail, interruptions, noisy audio, clarification, read-back, transfer, and hang-up without inventing missing details.	Must

ID	Capability	Requirement	Priority
FR-06	AI disclosure and consent	Provide configurable disclosure, recording notice/consent, and channel-specific compliance behavior before sensitive interaction.	Must
FR-07	Term extraction	Capture rate, accessorials, equipment, pickup/delivery windows, driver availability, and conditions with field-level confidence and evidence.	Must
FR-08	Negotiation	Use customer-approved strategy, ceiling, step size, concessions, and timeout; never reveal confidential ceiling or competing carrier identity.	Must
FR-09	Compliance verification	Check identity, operating authority, insurance, safety and internal blocks using approved sources immediately before booking.	Must
FR-10	Fraud controls	Detect suspicious contact or banking changes, mismatched identities, duplicate identities, spoof indicators, and anomalous booking patterns.	Must
FR-11	Offer comparison	Rank valid offers by total cost, service history, operational fit, compliance status, and customer preferences; show explanations.	Must
FR-12	Atomic booking	Guarantee one committed carrier per load version via idempotency key, lock, final policy check, and compensating rollback.	Must
FR-13	Confirmation	Generate and send rate confirmation from approved terms; require the configured e-sign/acknowledgment before final status.	Must
FR-14	TMS synchronization	Write selected carrier, rate, terms, timestamps, evidence links, and status; reconcile failed or delayed writes.	Must
FR-15	Human queue	Provide prioritized exceptions with recommended action, evidence, risk, countdown, and one-click approve/counter/reject.	Must
FR-16	Admin operations & control	Give the Supply Chain Strategic Operations Manager full authorized operational control: manage users, roles, integrations, configurations, policies, thresholds, workflows, exceptions, approvals, overrides, booking reconciliation, reporting, and pause/resume controls by load, customer, lane, agent, channel, or globally. Protected compliance and security controls remain enforced.	Must
FR-17	Learning capture	Collect corrections and downstream outcomes with reason codes; support approval before any policy or model update.	Should

ID	Capability	Requirement	Priority
FR-18	Analytics	Report funnel, automation, rate, time, compliance, model quality, carrier experience, service, and unit economics.	Must
FR-19	Multilingual	Support evaluated Spanish conversations and localized disclosures.	Could
FR-20	Load recovery	Re-source automatically after a carrier fall-off when customer policy and timing permit.	Should

Non-functional requirements

Area	Requirement
Availability	99.9% monthly orchestration availability after GA; degraded mode is observable and documented.
Latency	P95 <2 sec for agent response after end-of-turn; compliance pre-book checks complete within 10 sec under normal conditions.
Security	Encryption in transit/at rest, SSO/SAML, RBAC, tenant isolation, secrets management, least-privilege tool scopes, annual penetration test.
Privacy	Configurable retention; PII minimization/redaction; consent and deletion workflows; no customer data used for cross-customer training without explicit agreement.
Auditability	Immutable event log for prompts, model/tool versions, policy version, inputs, outputs, approvals, booking state, and evidence.
Scale	Initial 100 concurrent conversations and 100,000 eligible loads/month; architecture supports 10x without redesign.
Recovery	RPO ≤5 min and RTO ≤60 min for orchestration metadata; at-least-once events with idempotent consumers.

WEEK 2

Prioritization & Roadmap

PRIORITIZATION

Agentic workflow components

ID	Component	Responsibility	Key dependencies
C1	Load Intake & Context	Validate/normalize load, customer policy, and current version	TMS connector, schema service
C2	Policy & Eligibility	Apply hard constraints and calculate allowed autonomy	Rules engine, permissions
C3	Carrier Retrieval & Ranking	Find candidates and predict fit/acceptance	Carrier DB, ranking model
C4	Conversation Agents	Conduct bounded voice/SMS/email interaction	Telephony, ASR/TTS, LLM
C5	Negotiation	Extract offers and execute approved concession policy	Offer ledger, policy engine
C6	Compliance & Fraud	Verify carrier and block unacceptable risk	FMCSA/approved providers, internal data
C7	Selection & Booking	Select, atomically commit, confirm, and sync	Scoring, transaction service, TMS
C8	Human Control Plane	Escalate, review, approve, pause, and override	Ops console, RBAC
C9	Observability & Learning	Measure, trace, evaluate, and capture outcomes	Event store, eval pipeline

Component-level risk assessment

Component	Risk	Severity	Controls	Owner
C1	Stale or incomplete load	High	Schema validation, version lock, change events, invalidation, source timestamp	Product/Integrations
C2	Policy conflict or unauthorized override	Critical	Typed rules, test suite, approval workflow, versioning, fail closed	Product/Compliance
C3	Biased/poor ranking excludes viable carriers	High	Reason codes, coverage metrics, exploration cap, segment audits	ML/Product
C4	Misheard or hallucinated term	Critical	Field confidence, evidence span, read-back, alternate channel, human gate	AI/Operations
C4	Unlawful or unwanted outreach	Critical	Approved contacts, consent basis, time windows, disclosure, suppression list	Legal/Compliance

Component	Risk	Severity	Controls	Owner
C5	Rate above authority or confidential data leakage	Critical	Hard ceiling in external service, constrained tool, transcript tests	Product/Security
C6	Fraudulent carrier passes checks	Critical	Multi-source identity checks, freshness limits, anomaly rules, manual review	Compliance/Security
C7	Double booking or incorrect confirmation	Critical	Atomic lock, idempotency, recheck, transactional outbox, reconciliation	Engineering
C8	Slow or overloaded exception queue	High	Priority/SLA, staffing forecast, alerting, safe pause	Operations
C9	Feedback loop learns wrong behavior	High	Curated labels, delayed outcomes, approvals, holdout evaluation	ML/Product

AI suitability checklist

Check	Result	Rationale
Is AI necessary?	PASS	Speech and message interpretation, adaptive dialogue, and exception classification are unstructured; deterministic rules remain preferable for hard gates.
Do we have data?	CONDITIONAL	Pilot requires representative historical load, carrier, outreach, booking, compliance, and service data with usable timestamps and permissions.
Can the task be evaluated?	PASS	Critical facts and actions can be compared with transcripts, TMS records, approvals, and downstream outcomes.
Can accuracy meet customer needs?	CONDITIONAL	Noncritical conversational flexibility is acceptable; critical fields require read-back, high confidence, and deterministic validation.
Can it scale?	PASS	Parallel agents and stateless inference can scale horizontally; telephony, TMS, and human queue capacity are constraints.
How fast is feedback?	MIXED	Call understanding and booking accuracy are immediate; service and claims outcomes arrive days or weeks later.
What laws/policies apply?	REVIEW	FMCSA authority/insurance, TCPA/AI voice, call recording, privacy, contract, and customer communication rules require counsel and customer configuration.
Bias/fairness risk?	MATERIAL	Accent, dialect, language, carrier size, and sparse lane history can produce unequal interaction or ranking quality.
Transparency?	PASS WITH DESIGN	Expose extracted terms, evidence, policy checks, ranking factors, tool timestamps, and decision traces.

Check	Result	Rationale
Easy to judge good/bad?	PASS	Use structured term accuracy, policy compliance, booking match, service outcome, and human correction labels.

Prioritized MVP stories

Rank	Story	Priority rationale	Release
1	As an ops manager, I can define and test hard booking policies.	Removes unacceptable autonomy risk and unlocks every downstream action.	MVP
2	As a rep, I can launch voice/SMS/email outreach from a valid TMS load.	Creates the core workflow and measurable funnel.	MVP
3	As compliance, I can block booking when identity, authority, insurance, or internal rules fail.	Required before any live commitment.	MVP
4	As a rep, I receive structured offers with evidence and confidence.	Prevents hidden interpretation error and supports shadow mode.	MVP
5	As a system, I can select and commit one carrier atomically.	Completes the job and prevents double booking.	MVP
6	As a supervisor, I can review exceptions and pause automation.	Enables safe operations and incident response.	MVP
7	As a product team, I can run offline and live evaluations by segment.	Supports evidence-based go/no-go decisions.	MVP
8	As a customer, I can recover a fallen-off load automatically.	High value but depends on reliable initial booking.	MVP+1
9	As a Spanish-speaking dispatcher, I can complete the interaction in Spanish.	Expands coverage only after segment-specific quality is proven.	MVP+1

ROADMAP

Release	Scope	Duration	Exit criterion
R0 — Discovery & replay	Data audit, policy model, call/message replay, TMS and compliance sandbox, baseline	Weeks 1–4	Coverage baseline and critical-field dataset approved
R1 — Shadow MVP	Candidate generation, outbound agent, extraction, negotiation suggestions, compliance read-only, human books	Weeks 5–8	≥98% critical-field accuracy; zero severe safety test failures
R2 — Assisted booking	Human approval, rate confirmation, TMS write, exception console, 1–2 pilot lanes	Weeks 9–12	≥99.5% booking match; compliance evidence on 100% of bookings

Release	Scope	Duration	Exit criterion
R3 — Limited autonomy	Touchless booking for low-risk approved cohort; live guardrails and kill switch	Weeks 13–16	Launch criteria met for 4 consecutive weeks
R4 — Scale	More lanes/customers, recovery, Spanish pilot, improved ranking and capacity controls	Quarter 2	Unit economics and service non-inferiority sustained

WEEK 3

Implementation & Evaluation

IMPLEMENTATION PLAN

Technical architecture and control boundaries

Layer	Responsibilities	Control boundary
Experience	Ops console, supervisor dashboard, carrier voice/SMS/email	Role-based views; explicit approvals and pause controls
Orchestration	Workflow state machine, agent planner, timers, retries, load locks	Only allowlisted actions; per-agent tool scopes; bounded steps
Intelligence	ASR/TTS, LLM extraction/dialogue, carrier ranking, anomaly detection	Structured output; confidence thresholds; no direct transactional writes
Policy	Eligibility, contact, negotiation, compliance, approval, retention	Deterministic, versioned, testable, customer-specific
Transaction	Offer ledger, selection, booking, confirmation, TMS sync	Idempotency, final recheck, atomic state, reconciliation
Data & observability	Event log, evidence store, metrics, evals, model registry	Tenant isolation, access logging, retention, immutable audit trail

Integration contracts

System	Read	Write	Failure behavior
TMS	Load, customer, lane, status, carrier master	Offer/booking/status, notes, document link	Queue + retry; booking not final until reconciled
Telephony/messaging	Call status, audio/transcript, delivery status, opt-out	Calls, messages, transfers	Retry alternate channel within policy; suppress on opt-out
FMCSA / compliance provider	Authority, insurance, safety identifiers and timestamps	None	Fail closed for autonomous booking if required data is unavailable or stale
Identity/fraud provider	Phone/email/company match, risk signals	Case result where permitted	Route elevated risk to compliance
Document/e-sign	Acknowledgment/status	Rate confirmation	Hold final confirmation and alert queue

Evaluation strategy

Ground truth will be built from adjudicated historical interactions and controlled pilot data. Each critical label requires two trained reviewers; disagreements go to a senior operations or compliance adjudicator. Split by carrier and lane to reduce leakage, and maintain a frozen holdout set for release decisions.

Evaluation	Dataset / method	Metric	Release threshold
Load and policy understanding	1,000 representative loads with boundary cases	Field exact match; policy decision agreement	≥99.5% critical fields; 100% hard-rule tests
Speech robustness	500 calls stratified by noise, accent, device, interruption, voicemail	Word/semantic error; critical entity F1	Critical entity F1 ≥98%; no segment >3 pts below overall
Offer extraction	1,500 adjudicated conversations/messages	Field accuracy, unsupported extraction rate	≥99% critical terms; unsupported critical term <0.1%
Negotiation behavior	5,000 simulated/adversarial scenarios + replay	Ceiling breaches, disclosure leakage, policy adherence	Zero ceiling or restricted-data breaches
Compliance decision	2,000 carrier snapshots with fraud and stale-data cases	Recall on disqualifying conditions; evidence completeness	≥99.5% recall; 100% evidence bundle
Booking transaction	Concurrency, retry, duplicate, stale-load fault injection	Double booking; state mismatch	Zero double bookings; ≥99.9% reconciled within 5 min
Live business outcome	Controlled cohort by lane/time/customer	Touchless rate, time, margin, service, escalations	Targets met; guardrails non-inferior for 4 weeks

Model requirements

Criterion	Requirement	Rationale
Deployment	Enterprise API or privately hosted; contractual data controls and tenant isolation	Customer and carrier data require clear processing boundaries.
Modalities	Text and audio pipeline; streaming speech support	Real-time calls plus SMS/email.
Structured output	Schema-constrained JSON with tool calling	Workflow actions and evaluations require typed outputs.
Latency	P95 <2 sec end-of-turn; interruption handling	Long pauses damage calls and increase abandonment.
Context	At least 32K effective tokens plus retrieval	Load, customer, policy, carrier, and conversation context without oversized prompts.
Accuracy	Best available quality for extraction/dialogue; model cascade allowed	Critical term errors have financial and service consequences.
Observability	Model/version IDs, token and latency metrics, safety signals, deterministic replay inputs	Required for audit, debugging, and controlled rollout.
Customization	Prompt/RAG first; fine-tune only after error taxonomy proves durable gain	Reduces data and maintenance burden during MVP.
Fallback	Secondary model or deterministic script for critical degraded scenarios	Limits single-provider and model-regression risk.

EVALUATIONS — Helpful, Honest, Harmless

Dimension	Representative tests	Failure examples	Required action
Helpful	Covers requested load, communicates clearly, resolves normal objections, selects viable next step	Loops, long silence, irrelevant detail, premature transfer	Clarify, retry once, switch channel or escalate
Honest	Uses only current load/policy/tool data; discloses AI; distinguishes known, inferred, and unavailable data	Invented appointment, false compliance claim, hidden AI identity, fake competing bid data	Block action, flag severe incident, preserve trace
Harmless	Respects contact/recording/opt-out rules; protects confidential data; avoids manipulation and discrimination	Calls suppressed number, reveals ceiling, coercive claim, books blocked carrier	Immediate stop; incident workflow; customer notification per plan

Launch plan and criteria

Stage	Traffic / autonomy	Go criteria	Rollback trigger
Offline replay	No live contact	Data and policy coverage; critical eval thresholds met	Any severe unmitigated scenario
Shadow	Live context; agent output hidden from carrier; 0% booking writes	≥98% critical term accuracy; stable latency; adjudicated gap list	Privacy/security issue or repeated unsafe recommendation
Assisted beta	1–5% eligible loads; every booking approved	≥99.5% booking match; 100% compliance evidence; queue SLA met	Any hard-gate bypass or double booking
Limited autonomy	5–20% eligible low-risk loads	North-star ≥40% within cohort; rate/service non-inferior; severe incidents = 0	Guardrail threshold breach for 2 intervals or one critical incident
General availability	Progressive customer/lane expansion	North-star ≥60%; SLOs and unit economics sustained for 4 weeks	Auto-pause by customer/lane; global kill switch for systemic issue

Experiment design: matched or randomized rollout within comparable customer-lane-equipment-time strata. Analyze intent-to-treat and eligible-load cohorts. Pre-register primary metrics, non-inferiority margins, minimum sample size, and stopping rules with analytics before live autonomy.

WEEK 4

Data, Prompts & Responsible AI

DATA REQUIREMENTS

Data domain	Minimum fields	Purpose	Quality / governance
Load	Lane, stops, windows, equipment, commodity, weight, customer, rate target/ceiling, status/version	Context, eligibility, negotiation, booking	Required-field validity; source timestamp; versioning
Carrier	Legal name, USDOT/MC, contacts, equipment, lanes, preferences, service history	Retrieval, identity, ranking	Identity match; deduplication; contact permissions
Compliance	Authority, insurance, safety/internal blocks, provider, checked-at time	Hard booking gates	Freshness SLA; source evidence; fail-closed policy
Interactions	Channel events, audio where permitted, transcript, speaker turns, disclosures, opt-outs	Agent context, QA, evals	Consent; redaction; retention by customer/state policy
Offers	Rate, currency, accessorials, terms, confidence, evidence, expiry	Comparison and booking	Immutable ledger; critical-term confirmation
Outcomes	Accepted, booked, pickup/delivery, fall-off, claim, margin, corrections, reason codes	Business evaluation and learning	Delayed labels; attribution rules; reconciliation
Policy	Rule, scope, priority, effective dates, owner, tests, approval	Deterministic control	Versioned; four-eyes approval for critical rules

Training and evaluation data plan

Item	Decision
Fine-tuning	Not required for MVP. Use prompt/RAG, structured tools, and deterministic rules. Fine-tune only if a stable, high-volume error class remains after prompt and workflow fixes.
Initial quantity	Planning minimum: 10,000 historical load outcomes; 2,000 compliant interaction transcripts/messages; 1,500 adjudicated offer extractions; 2,000 compliance snapshots; 5,000 simulated negotiation cases.
Preparation	Deduplicate, redact unnecessary PII, align timestamps, label critical facts/evidence, stratify by lane/equipment/noise/accent/outcome, and document provenance/permission.
Iterative collection	Sample 100% of automated bookings for deterministic checks; risk-weight conversations for human QA; maintain random sample to detect unseen drift.
Knowledge base	Customer SOPs, load/customer rules, carrier communication preferences, accessorial definitions, approved scripts, escalation playbooks, and integration schemas.

Item	Decision
Update control	No autonomous production prompt, policy, or model changes. Every candidate release passes regression evals, approval, canary, and rollback readiness.

Prompt strategy

- Separate prompts by role: conversation, extraction, exception classification, and summary. Do not use one general-purpose prompt for the full workflow.
- Inject the minimum necessary load and policy context; retrieve customer-approved definitions and playbooks with source IDs and effective dates.
- Require schema-constrained outputs, explicit confidence, evidence spans, and an “unknown” state. Reject malformed or unsupported outputs.
- Use deterministic tool results for current facts. The agent cannot claim completion until the tool returns success.
- Constrain negotiation through tools: propose_offer(amount, terms) validates permissions and ceilings; the prompt never receives a needlessly precise confidential ceiling.
- Defend against prompt injection from transcripts/messages by treating carrier content as untrusted data and keeping instructions out of the conversation payload.
- Evaluate prompt changes against frozen regression, adversarial, and segment suites before canary release.

RESPONSIBLE AI RISKS & MITIGATION

Principle	Risk	Mitigation / requirement	Evidence
Accountability	Responsibility becomes unclear across AI, broker, and vendors	Named owner per policy/action; approval matrix; incident severity and review process	Decision log, approval, runbook, postmortem
Transparency	Carrier mistakes an AI agent for a person or cannot understand a decision	Configurable AI disclosure; explain critical terms and next steps; offer human transfer	Disclosure event, transcript, transfer outcome
Fairness	Lower quality for accents, dialects, Spanish, small carriers, or sparse-history lanes	Segmented evals; confidence-based handoff; alternate channel; ranking feature review	Parity dashboard and error taxonomy
Privacy	Calls expose PII or are retained beyond purpose	Minimize/redact, consent/notice, role access, customer retention schedule, deletion workflow	Access logs, retention jobs, deletion audit
Reliability	Hallucinated terms, stale data, tool failure, or double booking	Evidence, read-back, freshness checks, fail closed, idempotency, human queue, kill switch	Eval results, tool timestamps, transaction trace
Security/fraud	Spoofed identity, account takeover, data exfiltration, prompt injection	Multi-factor business identity signals, anomaly detection, least privilege, input isolation	Risk score, blocked action, security telemetry
Human agency	People cannot correct or stop the system	Pause/override at load/customer/global level; correction reason; no retaliation for transfer/opt-out	Control action log and response time

Regulatory and policy controls

FMCSA systems provide public carrier identity, safety, operating-authority, and licensing/insurance information [R2–R4]. Those records are evidence inputs, not a complete fraud guarantee; Shipt must also apply customer policy and identity/anomaly controls. The FCC has confirmed that TCPA restrictions on artificial or prerecorded voice encompass current AI-generated voices [R5]. Before live outreach, counsel must approve the contact basis, disclosure, consent, calling windows, suppression handling, recording rules, data retention, and state-specific configuration.

Control	Default product behavior
AI identity	Disclose that the caller is an automated AI assistant at the start, with a human-transfer option.
Contact authorization	Call/message only approved business contacts with documented lawful basis and customer policy.
Opt-out	Recognize verbal/text opt-out, confirm once, suppress promptly across channels, and log the action.
Recording	Use jurisdiction/customer configuration; capture notice/consent evidence; disable storage when not authorized.
Commitment authority	Disclose material load terms; book only within explicit broker authority and contractual workflow.
Data usage	No cross-customer model training from private data without explicit contractual permission.

Incident response

Severity	Example	Immediate response	Notification / follow-up
SEV-0	Double booking, hard compliance bypass, material unauthorized commitment, sensitive-data disclosure	Global or scoped kill switch; contain; preserve evidence	Customer + legal/security per plan; executive review; RCA before restart
SEV-1	Repeated critical-term error or unlawful contact pattern	Pause affected model/channel/cohort; manual recovery	Customer ops within agreed SLA; fix + regression suite
SEV-2	Latency, isolated failed confirmation, queue SLA miss	Retry/reconcile or route to human	Operational reporting and trend review

WEEK 5

Pricing & Business Case

PRICING

Cost and accuracy trade-offs

Decision	Choice	Benefit	Trade-off / guardrail
Model cascade	High-quality model for negotiation/critical extraction; smaller model for summarization/classification	Controls cost without weakening critical steps	Route by risk; compare on same evals
Parallelism	Bounded carrier batches, expanded only when needed	Faster coverage	Higher contact cost and carrier fatigue; frequency caps
Compliance freshness	Recheck immediately before booking; cache only within approved TTL	Reduces API cost and latency	Fail closed if stale/unavailable
Audio retention	Configurable; retain transcript/evidence by default, audio only with permission and need	Lower privacy and storage risk	May reduce debugging evidence
Fine-tuning	Defer until a stable error class proves ROI	Faster iteration and lower governance cost	Potentially higher inference cost initially

Development costs — directional planning range

Workstream	Indicative team / spend	Range or duration
Product, design, operations	1 PM, 0.5 designer, 1 operations/domain lead	6–9 months
Engineering	3 backend/platform, 2 integrations, 1 frontend, 1 SRE/security	6–9 months
AI/ML and evaluation	2 ML/AI engineers, 1 data engineer, 1–1.5 QA/eval	6–9 months
Legal/compliance and pilot enablement	Fractional counsel/compliance, implementation support	\$100K–\$300K external/allocated
Infrastructure/vendor sandbox and test traffic	Cloud, telephony, models, compliance APIs, observability	\$75K–\$250K
Total one-time build estimate	Fully loaded labor plus external spend	\$1.5M–\$3.0M planning range

Ranges are planning assumptions. Finance must replace them with company rates, vendor quotes, security/compliance requirements, and the actual pilot integration scope.

Operational costs

Cost driver	Primary unit	Planning range / control
Telephony and messaging	Per minute / message	Instrument by channel; control with batching, timeout, and carrier preference

Cost driver	Primary unit	Planning range / control
Speech and model inference	Per audio minute / token / turn	Use risk-tier routing, prompt compression, caching of non-sensitive static context
Compliance and fraud data	Per lookup / carrier / month	Recheck at booking; approved freshness-aware cache
Cloud, eventing, storage, observability	Per conversation / retained artifact	Tiered retention, redaction, sampling without losing critical audit records
Human exception operations	Minutes per escalated load	Top unit-economics driver; reduce through root-cause fixes, not lower standards
Composite variable cost hypothesis	Per eligible load / booked load	\$1.50–\$4.00 per eligible load; validate in pilot
Fixed run cost hypothesis	Per month at initial scale	\$50K–\$150K excluding customer implementation and sales

Market size — bottom-up planning model

Because public industry categories do not cleanly isolate the target segment and compatible load volume, use a bottom-up model tied to verified customer counts and actual processed loads. Do not present the scenario below as an external market forecast.

Scenario	Planning assumptions	Annual recurring revenue potential
TAM hypothesis	5,000 target brokers × 2,000 booked loads/month × \$10/load	\$1.20B variable revenue, plus platform fees
SAM hypothesis	1,500 integration-ready brokers × 2,000 booked loads/month × \$10/load	\$360M variable revenue, plus platform fees
3-year SOM hypothesis	50 customers × 2,000 booked loads/month × \$10/load	\$12M variable revenue, plus platform fees

Validation inputs: current active brokerage population, distribution of monthly load volume, TMS compatibility, eligible-load share, willingness to pay, sales cycle, implementation capacity, churn, and pricing response.

Revenue potential and pricing models

Model	Strength	Risk	Recommendation
Per booked load	Aligns price with realized workflow value	Can discourage use on low-value loads; disputes over attribution	Primary variable fee with clear booking definition
Per eligible/attempted load	Recovers outreach and inference cost	Customer pays even when not covered	Use only as overage or lower-priced usage fee
Platform subscription	Predictable revenue; covers integrations/control plane	May be expensive for low-volume customers	Base fee tiered by volume and features
Share of savings / margin	Strong outcome alignment	Baseline and attribution are contentious	Optional enterprise pilot with audit-ready baseline
Per seat	Familiar procurement	Misaligned with automation and shrinking seat needs	Avoid as primary metric

Directional pricing

Recommended hypothesis \$5,000–\$15,000 monthly platform fee per brokerage environment, plus \$8–\$15 per successfully booked load and pass-through premium data/telephony where material. Offer an implementation fee for nonstandard TMS work. Validate with 8–12 design partners and a paid pilot.

Scenario	Customer volume	Illustrative annual revenue
Starter mid-market	1,000 booked loads/month; \$5K platform + \$12/load	\$204K
Core	2,500 booked loads/month; \$8K platform + \$10/load	\$396K
Enterprise	10,000 booked loads/month; \$15K platform + \$8/load	\$1.14M

Unit-economics decision rule

Proceed to broad GA only if contribution margin is $\geq 65\%$ at the expected customer mix, exception handling remains below 5 human minutes per eligible load, and the customer's measured value—labor capacity, faster coverage, margin, or service improvement—supports at least a 3 $\times$ value-to-price ratio. Replace assumptions with observed pilot cohorts before pricing approval.

Acceptance Criteria & Operating Readiness

MVP acceptance criteria

- A valid in-scope TMS load can complete intake, candidate generation, approved outreach, offer extraction, compliance recheck, atomic booking, confirmation, and TMS reconciliation.
- Every booking has the exact load version, selected carrier identity, agreed rate/terms, policy version, compliance timestamps, conversation evidence, and transaction trace.
- The system cannot book above the ceiling, with an ineligible carrier, after the load changes materially, or when a required tool is unavailable/stale.
- The Supply Chain Strategic Operations Manager can perform all authorized operations, including configuration, user and policy management, workflow intervention, approvals, overrides, reconciliation, reporting, and scoped or global pause/resume controls.
- Opt-out and recording behavior follows configured policy and is testable end to end.
- All launch thresholds are met for the defined cohort and interval; rollback has been rehearsed.

Operational readiness checklist

Area	Ready when	Accountable
Product	Cohort, eligibility, metrics, approval matrix, and customer configuration signed off	Product
Engineering	SLOs, load tests, fault injection, reconciliation, backups, and rollback verified	Engineering/SRE
AI/ML	Frozen eval suite passed; segment gaps documented; model/prompt registry and rollback active	AI/ML
Security/privacy	Threat model, access review, vendor review, pen test, retention/deletion controls complete	Security/Privacy
Legal/compliance	Outreach, AI disclosure, recording, FMCSA data use, contracting, and incident notices approved	Legal/Compliance
Operations	Queue staffed; playbooks, training, escalations, manual fallback, and customer communications rehearsed	Operations/Customer Success
Analytics/Finance	Baseline, experiment design, cost telemetry, contribution margin, and customer ROI dashboard approved	Analytics/Finance

Open questions

Question	Why it matters	Resolution method
Which TMS and compliance providers define the first integration?	Determines technical scope and data freshness	Select pilot and run API/data audit
What is the actual eligible-load share?	Controls automation ceiling and business case	Replay 8–12 weeks of load history

Question	Why it matters	Resolution method
Which carrier contacts have approved outreach and recording basis?	Controls legal risk and reachable supply	Legal review + contact provenance audit
How should total cost and service trade off in carrier selection?	Determines ranking and explainability	Historical analysis + customer policy workshop
What exceptions consume the most human time?	Sets roadmap and eval weighting	Shadow operations and coded exception study
What outcome justifies willingness to pay?	Determines packaging and sales proof	Paid design-partner interviews and pilot ROI

References

R1 — Bureau of Transportation Statistics, “Moving Goods in the United States.” [Source](#)

R2 — FMCSA SAFER Web, Company Snapshot and safety information. [Source](#)

R3 — FMCSA Licensing & Insurance carrier search. [Source](#)

R4 — FMCSA Analysis & Information system and registration/safety data. [Source](#)

R5 — FCC, “FCC Confirms that TCPA Applies to AI Technologies that Generate Human Voices,” February 8, 2024. [Source](#)

Regulatory citations support product requirements but do not replace legal advice. Requirements must be reviewed against the customer's contacts, call purpose, jurisdictions, contracts, and current law before production use.